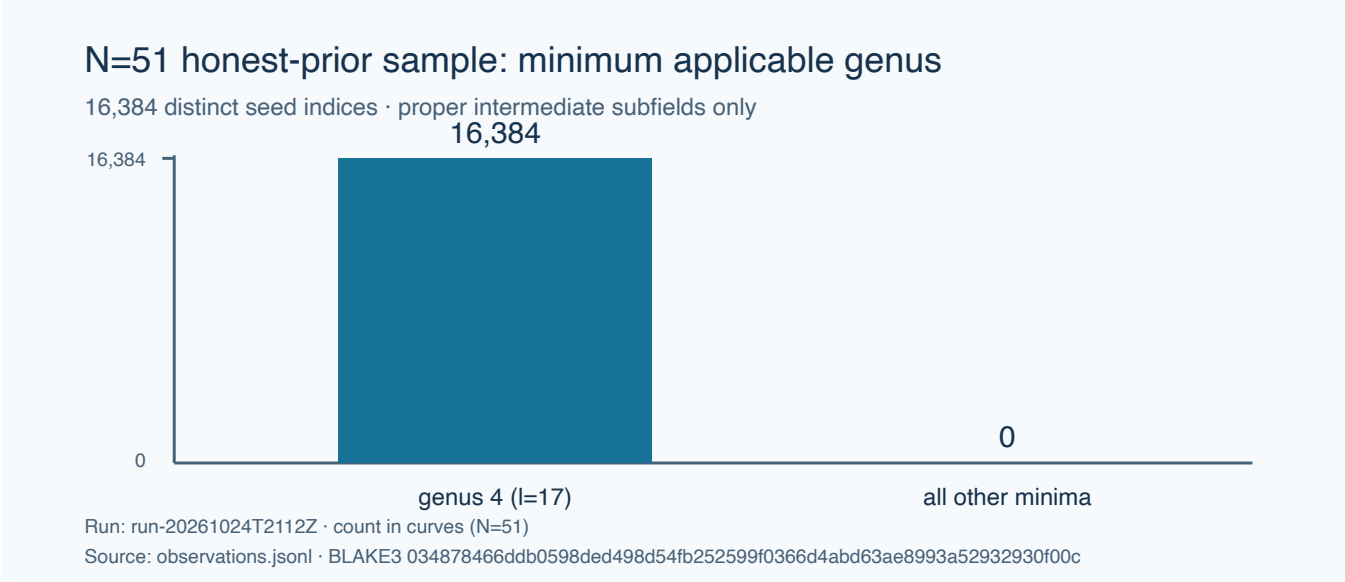
N=51 honest-prior GHS genus distribution

24 October 2026 · phase 1 structural diagnostic

The frozen protocol sampled 16,384 curves from public seed `20261024202610242026102420261024202610242026102420261024202610242026102420261024` in GF(2^51), polynomial $x^{51} + x^6 + x^3 + x + 1$. The BLAKE3 XOF generator and complete GHS audits are in the source repository. The statistic is the smallest genus with a descending row over a proper intermediate subfield ($l = 3$ or $l = 17$); the $l = 1$ audit remains in raw records but is excluded from this statistic.



Observed frequencies

Threshold	Hits / 16,384	Empirical p
$g \leq 0, 1, 2, 3$ (each)	0	0
$g \leq 4, 8$ (each)	16,384	1

For each zero-hit threshold, modeling the indexed BLAKE3 outputs as independent uniform draws, the exact one-sided 95% binomial upper bound is $1 - 0.05^{\frac{1}{16384}} = 0.0001828282722735608$. No finite empirical grind-bit estimate is assigned to a zero-hit event. The observed $g = 4$ row uses $l = 17$ for every index.

Scope and replay

This is a sampled structural observation. Hidden-route secrecy, detector error, destination solving, complete cost, and matched rho were not measured. The run used Darwin arm64 kernel 25.6.0 without CPU isolation; it makes no speed claim. Reproduce from base revision `0200e2b2af7c608b100930be9397124441c2710b` plus this PR with:

```
cargo run --release --example seed_grind_honest_prior -- 16384 <seed-hex> <new-output-dir>
cargo run --release --example seed_grind_honest_prior -- --verify <output-dir>
```

The scoreboard records the panel; the existing IC progress timeline and method-bound frontier remain unchanged because this phase adds no priced pipeline or method bound.

The verifier rederived and compared all 16,384 records and their aggregate successfully. The run directory `seed_grind_trapdoor_20261024_run/observations.jsonl` is 4,970,702 bytes, BLAKE3 `034878466ddb0598ded498d54fb252599f0366d4abd63ae8993a52932930f00c`. See the run directory's `summary.json`, `observations.jsonl`, this report's `REPORT.md` and the frozen section 11 of `RESEARCH_SEED_GRIND_TRAPDOOR_20261002.md` for the full seed, command, derivation and limits. Replay used the same implementation and host; independent cross-implementation verification remains open.

Next discriminating phase

Freeze seed freedom and a detector rule before selecting curves, then measure success and detector errors with matched controls. Keep any subsequent end-to-end comparison separately accounted.